TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Substitution: get a variable alone, put its expression into the other equation, solve, substitute back.
Variable drops out? False statement (like 0 = 7): no solution. True statement (like 5 = 5): infinite.
Check: put both values into both equations.

1. Which point is the solution of $y = -x + 4$ and $y = x - 2$?

Tip: check the point in both

A. (3, 1)

B. (1, 3)

C. (4, −2)

2. Solve $4x + y = 9$ for y.

Tip: undo + $4x$ by subtracting

F. $y = 4x + 9$

G. $y = -4x + 9$

H. $y = 9 - x$

3. What is y when $x = 3$ in $y = 2x - 1$?

> **Tip:** substitute 3 for x

A. 7

B. 6

C. 5

4. Which system has no solution?

> **Tip:** same slope, different b: none

F. $y = x + 1$ and $y = 2x + 1$

G. $y = x + 1$ and $y = -x + 1$

H. $y = x + 1$ and $y = x - 3$